# DAA Lab Viva Questions and Simple Answers

Tag key: PDF means the detail appears in the study guide. General DAA means it is standard algorithm knowledge added for revision.

## 1. Recursive tree traversals

Q: What is a binary tree?

A: A tree where each node has at most two children: a left child and a right child.

Q: What is tree traversal?

A: Visiting every node of a tree in a particular order.

Q: What is inorder traversal?

A: Visit left subtree, node, then right subtree: Left - Node - Right.

Q: What is preorder traversal?

A: Visit the node, left subtree, then right subtree: Node - Left - Right.

Q: What is postorder traversal?

A: Visit the left subtree, right subtree, then node: Left - Right - Node.

Q: How does recursion stop?

A: When the current node is null, return.

Q: What does inorder print for the PDF's sample tree?

A: 4 2 5 1 3. PDF

Q: What does preorder print for the sample tree?

A: 1 2 4 5 3. PDF

Q: What does postorder print for the sample tree?

A: 4 5 2 3 1. PDF

Q: What is the time complexity of each traversal?

A: O(n), because each of the n nodes is visited once. General DAA

Q: Does inorder always print values in sorted order?

A: No. It does so only when the tree is a binary search tree.

## 2. Iterative tree traversals

Q: What does "iterative" mean?

A: The traversal uses loops and an explicit data structure instead of recursive calls.

Q: Which data structure is used for iterative tree traversal?

A: A stack. PDF

Q: Why is a stack useful here?

A: It remembers which nodes still need to be visited.

Q: How is iterative inorder traversal done?

A: Push nodes while moving left; pop and visit a node; then move to its right child.

Q: Why push the right child before the left in iterative preorder?

A: A stack is last-in, first-out, so pushing right first makes left come out first.

Q: Do recursive and iterative traversals give different orders?

A: No. They can produce the same traversal order using different methods.

Q: What is their time complexity?

A: O(n). General DAA

## 3. Hashing to count quadruples

Q: What is a quadruple in this problem?

A: One selected value from each of four arrays.

Q: What does the program count?

A: The combinations whose four values add up to the target x.

Q: How does hashing help?

A: Store sums from two arrays in a hash table, then look up the complementary sum needed from the other two arrays.

Q: Why store a frequency for each sum?

A: Different pairs can have the same sum, and every matching combination must be counted.

Q: What sum is looked up for a pair with sum s?

A: x - s.

Q: What is the expected time complexity?

A: $O(n^2)$, using pair sums from two pairs of arrays. General DAA

Q: What is the extra space complexity?

A: $O(n^2)$ for the stored pair sums. General DAA

Q: Must the arrays be sorted for this hash method?

A: No. The hash method does not need sorted arrays. General DAA

Q: What output does the PDF give?

A: Count = 4. PDF

## 4. Graph traversal: BFS and DFS

Q: What is a graph traversal?

A: A systematic way to visit vertices in a graph.

Q: What does BFS stand for?

A: Breadth-First Search. The PDF also calls it Breadth First Traversal, or BFT.

Q: Which data structure does BFS use?

A: A queue.

Q: How does BFS work?

A: It visits the start vertex, then visits its neighbors level by level.

Q: What does DFS stand for?

A: Depth-First Search. The PDF also calls it Depth First Traversal, or DFT.

Q: Which data structure does DFS use?

A: A stack, or the programming language’s recursion stack.

Q: How does DFS work?

A: It follows one path deeply, then backtracks to explore other paths.

Q: Why do BFS and DFS need a visited set or array?

A: To avoid visiting the same vertex repeatedly, especially when the graph has cycles.

Q: Which traversal finds shortest paths in an unweighted graph?

A: BFS. General DAA

Q: What is the time complexity with an adjacency list?

A: O(V + E), where V is the number of vertices and E is the number of edges. General DAA

Q: What BFS output does the PDF give from vertex 0?

A: 0 1 2 3 4. PDF

Q: Is the PDF's DFS example consistent with its listed edges?

A: No. Its listed edges do not include 2 -> 0, but its DFS example starts 2 0 1 3. Also, the example omits other reachable vertices. Treat it as a printed example, not a trace of those edges.

## 5. Heap sort

Q: What is a heap?

A: A complete binary tree that follows a parent-child ordering rule.

Q: What is a max-heap?

A: A heap where each parent is greater than or equal to its children.

Q: Which heap does the PDF mention for heap sort?

A: A binary max-heap. PDF

Q: How does heap sort work?

A: Build a max-heap, move the largest value to the end, then restore the heap and repeat.

Q: Why does a max-heap produce ascending order?

A: Each largest value is moved to the last available position.

Q: What is heapify?

A: Restoring the heap property by moving a node down as needed.

Q: What is heap sort's time complexity?

A: O(n log n) in the usual implementation. General DAA

Q: How much extra space does in-place heap sort use?

A: O(1) auxiliary space. General DAA

Q: Is heap sort stable?

A: No. Equal values may change their relative order. General DAA

Q: Does the PDF give a sample input or output for heap sort?

A: No. It gives the topic title and "Binary Max-Heap" only.

## 6. Assign mice to holes

Q: What is the goal of this problem?

A: Assign each mouse to a different hole so the longest travel time is as small as possible.

Q: What is the greedy method?

A: Sort the mouse positions and hole positions, then pair them in order.

Q: How is the final answer calculated?

A: Find the largest distance among all paired mice and holes.

Q: What happens if there are different numbers of mice and holes?

A: The PDF function returns -1. PDF

Q: What are the PDF's mouse and hole positions?

A: Mice: [4, -4, 2]; holes: [4, 0, 5]. PDF

Q: What is the expected answer for that example?

A: 4, the maximum distance after pairing the sorted positions. PDF

Q: What is the time complexity?

A: O(n log n) because of sorting. General DAA

Q: Is the objective to minimize the sum of distances?

A: No. It minimizes the maximum individual travel distance.

## 7. Minimum cost to cut a board into squares

Q: What is the goal?

A: Cut the board into 1-by-1 squares with the lowest total cost.

Q: What is the greedy choice?

A: Make the highest-cost available cut first. PDF

Q: Why make the highest-cost cut first?

A: A costly cut should cross as few existing pieces as possible.

Q: What multiplier applies to a horizontal cut?

A: The current number of vertical pieces.

Q: What multiplier applies to a vertical cut?

A: The current number of horizontal pieces.

Q: What values does the PDF use?

A: m = 6, n = 4, X = {2,1,3,1,4}, and Y = {4,1,2}. PDF

Q: What is the expected output?

A: 42. PDF

Q: What is the time complexity if the costs must be sorted?

A: O(m log m + n log n) for the two cost lists. General DAA

## 8. Connect N ropes with minimum cost

Q: What is the goal?

A: Connect all ropes into one while minimizing the total connection cost.

Q: How much does it cost to connect two ropes?

A: The sum of their lengths. PDF

Q: Which data structure is used?

A: A min-heap or min-priority queue. PDF

Q: What is the greedy step?

A: Remove the two shortest ropes, connect them, and put their combined length back in the heap.

Q: When does the algorithm stop?

A: When only one rope remains.

Q: Why use a min-heap?

A: It quickly provides the two shortest current ropes.

Q: What is the time complexity?

A: O(n log n) with a binary min-heap. General DAA

Q: Does the PDF give sample rope lengths or an expected output?

A: No. It explains the method and says it is similar to Huffman coding.

## 9. Rearrange a string so equal characters are d apart

Q: What is the goal?

A: Rearrange the string so identical characters are at least d positions apart.

Q: Which data structure does the PDF mention?

A: A max-heap. PDF

Q: What does the max-heap prioritize?

A: The character that still has the greatest frequency.

Q: Why is a cooldown queue useful?

A: It prevents a character from being used again before the required distance has passed. General DAA

Q: What does d = 2 mean?

A: Repeated characters must have at least one character between them.

Q: What should the program do if no valid character is available?

A: Report that the string cannot be rearranged.

Q: What input and example output does the PDF show?

A: Input: aaaabbbcc, with d = 2. One example output is abacabcab; the PDF also says to print “Cannot be rearranged” when no valid arrangement exists. PDF

Q: Is there only one valid output?

A: No. Several different rearrangements may be valid.

## 10. Minimization of cash flow

Q: What is the goal?

A: Settle debts among people using fewer transactions.

Q: What is a net balance?

A: Money owed to the person minus money the person owes.

Q: Who is a debtor in the net-balance array?

A: A person with a negative balance.

Q: Who is a creditor?

A: A person with a positive balance.

Q: What is the greedy step?

A: Settle between the person with the largest debt and the person owed the most.

Q: How much should they transfer?

A: The smaller of the debtor's debt and the creditor's amount due.

Q: What debts are in the PDF example?

A: Person 0 owes 1000 to person 1 and 2000 to person 2; person 1 owes 5000 to person 2. PDF

Q: What settlement does the PDF give?

A: Person 1 pays 4000 to person 2, and person 0 pays 3000 to person 2. PDF

Q: What convention does that example use for the matrix?

A: graph[i][j] means person i owes person j.

## 11. Greedy approximate K-centres

Q: What is the K-centres problem?

A: Choose k cities for warehouses so the farthest city is as close as possible to its nearest warehouse.

Q: What does k mean?

A: The number of cities selected as centers or warehouse locations.

Q: What is a common greedy approximate method?

A: Repeatedly select the city farthest from its nearest already selected center. General DAA

Q: Why is the method called approximate?

A: It may not find the exact best answer, but it has a quality guarantee under certain conditions.

Q: What guarantee does farthest-first give for metric distances?

A: A 2-approximation: its maximum distance is at most twice the optimal value. General DAA

Q: What is a metric distance?

A: A distance that is nonnegative, symmetric, zero from a point to itself, and satisfies the triangle inequality. General DAA

Q: What goal does the PDF state?

A: Select k cities from n cities to place warehouses, minimizing the maximum distance from any city to a warehouse. PDF

Q: Does the PDF give an example or expected output?

A: No. It gives the problem objective only.